

CHATHAM ISLAND, New Zealand

WMO: 939850

Lat: 43.8167S

Lon: 176.4750W

Elev: 40

StdP: 14.67

Time Zone: 12.75 (NZC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.4	39.1	30.8	25.1	40.7	32.5	27.0	41.5	40.8	47.9	36.7	46.6	8.9	180	0.750

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.8	69.5	63.7	67.6	62.7	66.0	61.7	65.1	67.8	63.9	66.2	62.9	65.0	17.2	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.9	89.2	66.3	62.9	86.0	65.2	62.0	83.2	64.3	30.1	67.9	29.3	66.5	28.5	65.1	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.1	31.4	28.3	33.9	71.5	2.2	2.5	32.3	73.2	31.0	74.7	29.8	76.1	28.2	77.9		
(5)			32.3	67.1	1.8	1.3	31.0	68.0	29.9	68.7	28.9	69.5	27.5	70.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	58.7	59.5	58.1	54.7	51.6	48.6	46.9	47.7	49.1	50.8	52.8	56.7	(6)
(7)		DBStd	5.75	3.86	3.79	3.87	4.09	3.68	3.85	3.61	3.42	3.65	3.61	4.02	4.29	(7)
(8)		HDD50	413	0	0	0	6	24	73	111	87	60	34	16	2	(8)
(9)		HDD65	4426	198	158	216	309	414	493	562	535	477	439	366	259	(9)
(10)		CDD50	1470	269	266	250	147	75	30	14	17	33	60	100	209	(10)
(11)		CDD65	8	2	4	1	0	0	0	0	0	0	0	0	1	(11)
(12)		CDH74	2	1	1	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	14.8	14.3	13.4	14.0	14.5	14.1	15.7	14.6	14.9	15.8	16.0	15.7	14.7	(14)
(15)	Precipitation	PrecAvg	35.3	2.3	2.4	3.1	3.2	3.7	4.1	3.0	3.1	2.6	2.5	2.2	2.5	(15)
(16)		PrecMax	42.3	6.7	4.8	7.0	6.0	7.1	6.5	5.9	6.1	4.9	4.8	7.5	6.5	(16)
(17)		PrecMin	25.7	0.4	0.4	0.6	1.3	0.4	2.6	1.6	0.7	0.4	0.9	0.6	0.7	(17)
(18)		PrecStd	4.6	1.5	1.2	1.6	1.3	1.4	1.0	1.1	1.4	1.2	1.1	1.5	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.4	72.4	70.1	65.9	62.0	58.7	57.0	57.7	59.6	62.7	66.7	70.6	(19)
(20)			MCWB	64.1	65.6	64.3	62.2	59.1	56.5	54.5	54.1	55.3	55.8	59.5	63.1	(20)
(21)		2%	DB	68.8	69.7	67.3	63.9	60.3	57.0	55.5	55.5	57.3	60.0	63.4	67.3	(21)
(22)			MCWB	63.2	64.3	62.9	60.8	58.3	55.2	53.6	53.3	54.1	55.4	57.8	62.0	(22)
(23)		5%	DB	66.9	67.5	65.5	62.3	59.1	55.9	54.1	54.2	56.0	58.2	61.1	65.0	(23)
(24)			MCWB	62.2	63.0	61.8	59.8	57.4	54.2	52.4	52.2	53.6	54.8	56.7	60.6	(24)
(25)		10%	DB	65.1	65.7	64.1	61.0	57.9	54.7	52.9	53.1	54.9	56.6	59.3	63.0	(25)
(26)			MCWB	61.0	62.1	61.1	58.8	56.1	53.0	51.2	51.2	52.9	53.8	55.9	59.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.1	67.3	65.6	63.4	60.4	57.4	55.5	55.3	56.5	57.8	61.0	64.8	(27)
(28)			MCDWB	69.6	70.4	68.1	64.9	61.2	57.9	56.3	56.1	58.0	60.1	65.5	67.9	(28)
(29)		2%	WB	64.5	65.5	64.1	61.7	58.9	55.8	54.0	54.0	55.2	56.6	59.2	63.1	(29)
(30)			MCDWB	67.3	68.1	65.9	63.0	59.8	56.7	54.9	55.0	56.3	58.5	61.9	65.7	(30)
(31)		5%	WB	63.3	64.2	63.0	60.5	57.8	54.6	52.8	52.8	54.3	55.5	57.9	61.7	(31)
(32)			MCDWB	65.7	66.5	64.6	61.8	58.9	55.7	53.9	53.9	55.5	57.3	60.1	64.0	(32)
(33)		10%	WB	62.1	62.9	61.9	59.2	56.4	53.3	51.5	51.5	53.2	54.5	56.6	60.3	(33)
(34)			MCDWB	64.3	65.1	63.6	60.6	57.7	54.6	52.8	52.7	54.5	56.2	58.7	62.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	9.8	8.9	8.0	7.8	6.9	7.3	7.3	8.0	8.9	9.6	9.7	(35)
(36)			MCDBR	12.2	11.6	10.3	9.1	9.0	7.6	8.3	8.5	9.2	11.0	11.7	11.4	(36)
(37)			MCWBR	8.2	7.6	7.2	7.3	8.1	7.0	7.5	7.1	7.5	8.2	8.1	7.5	(37)
(38)		5% WB	MCDBR	9.9	9.7	8.4	8.1	8.1	7.0	7.8	7.7	7.9	8.8	9.9	9.6	(38)
(39)			MCWBR	7.3	7.4	6.7	7.3	7.6	6.9	7.4	7.0	7.0	7.4	7.6	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.362	0.347	0.339	0.332	0.323	0.328	0.321	0.329	0.346	0.348	0.340	0.349	(40)	
(41)		taud	2.499	2.555	2.574	2.575	2.564	2.456	2.516	2.539	2.487	2.489	2.525	2.532	(41)	
(42)		Ebn at Noon	305	301	288	266	244	225	240	263	280	297	310	311	(42)	
(43)		Edh at Noon	35	31	28	24	20	20	21	24	30	33	34	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1982	1683	1246	804	513	386	466	701	1072	1494	1904	2015	(44)	
(45)		RadStd	131	115	61	42	30	23	29	49	64	99	125	124	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.03	N/A	+1.06	+1.84	+0.77	+0.83	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page